



Multi-Digit Multiplication Warm-Up Week

Grade 5 • Mathematics
Warm-Ups

Original Practice • Assessment • Complete Teacher Key



Teacher Guide

Level: Grade 5

State: North Carolina

Subject: Mathematics

Resource type: Warm-Ups

Estimated time: Five 10-minute warm-ups

Standard / objective: NC.5.NBT.5 - Demonstrate fluency with multiplication of whole numbers up to three digits by two digits using the standard algorithm.

Source note: North Carolina Department of Public Instruction, Mathematics Standards Vertical Progression:

<https://www.dpi.nc.gov/documents/publications/catalog/ma196-vertical-progression-math/open>; NC Check-Ins 2.0 Mathematics

Specifications: <https://www.dpi.nc.gov/documents/accountability/testing/nc-checkin/nc-check-ins-20-mathematics-grades-3-8-specifications> (accessed September 14, 2026).

What is included: branded cover, teacher guide, student directions, 16 original practice items, 5-item exit ticket, complete answer key/worked guidance, differentiation supports, printing notes, and license statement.

Purpose: Build accurate, fluent standard-algorithm multiplication with estimation and error analysis.

Preparation: Print student pages single-sided or assign through a secure LMS. Review the key and select tools allowed for your learners. No special commercial curriculum is required.

Access and differentiation

AIG: require a second representation, proof, counterexample, transfer task, or student-created extension. **504:** provide extended time, chunked sections, breaks, reduced copying, and accessible digital zoom as specified by the plan. **EC:** preteach vocabulary, model one item, provide manipulatives/visuals, and accept aligned response modes from the IEP. **ESL:** preview key terms, provide bilingual resources and sentence frames, and allow partner rehearsal. Use each learner's current plan; this guide does not replace it.



Student Practice

Learning focus: Build accurate, fluent standard-algorithm multiplication with estimation and error analysis.

Directions: Complete all items. Show calculations, cite evidence, or explain your reasoning when requested. Check your work before submitting.

1. 34×26

2. 128×14

3. 305×22

4. Estimate 247×31 using compatible numbers.

5. 46×19

6. 215×43

7. A printer makes 125 pages per minute for 24 minutes.

8. Find the error: $63 \times 27 = 1,071$.



Student Practice - Continued

9. 408×16

10. 72×35

11. Which is reasonable for 296×18 : 532, 5,328, or 53,280?

12. A theater has 36 rows of 128 seats.

13. 504×29

14. 99×47

15. Use a second method to check 214×32 .

16. Explain why the second partial product shifts one place.



Exit Ticket

Complete independently in 8-10 minutes.

1. State the main skill or concept in your own words.
2. Create one accurate example that uses today's skill.
3. Explain how you checked your example.
4. Identify one common error and correct it.
5. Rate your confidence from 1-4 and name your next practice step.

Quick rubric: 4 = accurate, complete, and well justified; 3 = accurate with minor omission; 2 = partial understanding; 1 = limited evidence; 0 = blank or unrelated.



Complete Teacher Answer Key

1. 884
2. 1,792
3. 6,710
4. About 7,500
5. 874
6. 9,245
7. 3,000 pages
8. Missing tens-place value; correct product 1,701
9. 6,528
10. 2,520
11. 5,328
12. 4,608 seats
13. 14,616
14. 4,653
15. 6,848; e.g., $214 \times 30 + 214 \times 2$
16. It represents multiplication by tens, not ones.

Exit ticket guidance: Responses vary. Award full credit when the student accurately explains the focus, supplies a correct example, describes a valid check, identifies and corrects a realistic misconception, and chooses a specific next step.

Verification: Fixed responses and calculations were checked during production. Open responses are evaluated against the criteria above.

Originality and use: This packet contains original educational content. It does not reproduce released-test items or branded curriculum. One purchaser may use it with their own students; redistribution, resale, and public posting are not permitted.